Subject outline

Lecture topics (by week)

1 – Introduction to quantum computing and maths basics
2 – Single qubit representations and logic operations
3 – Two qubit states and logic gates
4 – Multi-qubit states and quantum arithmetic
5 – Basic quantum algorithms
6 – Period finding, cryptography and quantum factoring
7 – Shor's algorithm, post-quantum crypto, quantum key distribution
8 – Quantum search algorithms
9 – Grover search applications, optimisation problems
10 – Solving optimisation problems on quantum computers
11 – Applications in quantum machine learning
12 – Real quantum computer devices

Assignment schedule:
#1: Hand out in Week 2
#2: Hand out in Week 8

Week 1

Lecture 1

1.1 A very brief history of computing

1.2 The quantum world and quantum computing

Lecture 2

2.1 The mathematics of quantum states

2.2 Complex numbers and quantum amplitudes

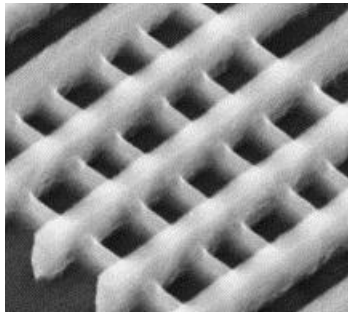
2.3 Basic linear algebra: ket and matrix notation

2.4 State representation in the QUI

Practice class 1

The Quantum User Interface (QUI), lecture 1 & 2 review exercises

Lecture 1 recap



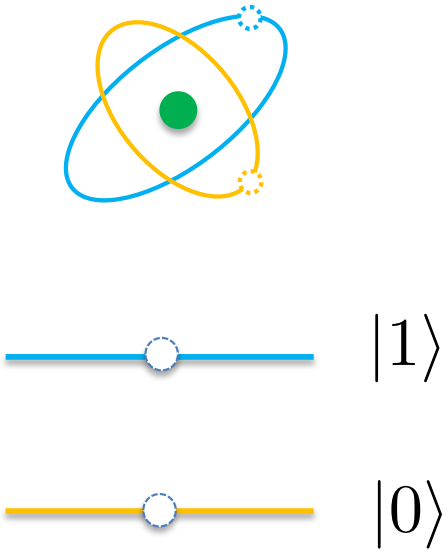
Intel.com

on “1”

off “0”

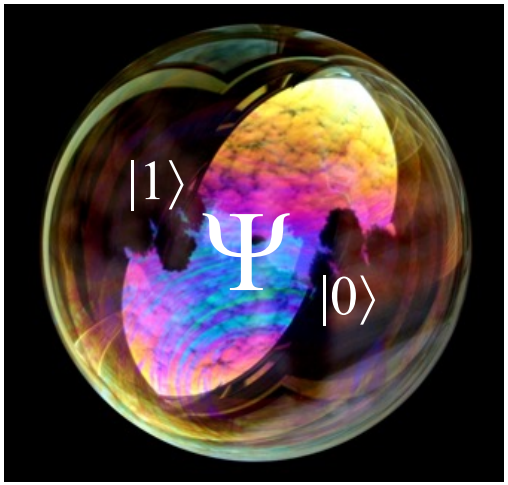
Only “0” <u>or</u> “1” at a time in any given physical representation (transistor)

Notation for bits:
Classical $\rightarrow$ 0 and 1
Quantum $\rightarrow$ $|0\rangle$ and $|1\rangle$



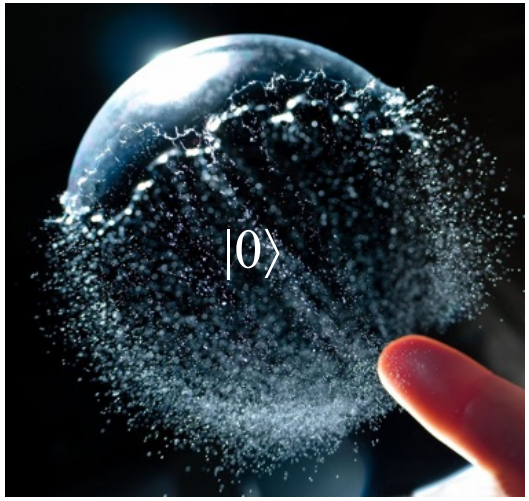
One electron in quantum superposition.

quantum superposition



$|0\rangle$ “and” $|1\rangle$

measurement/observation



state collapse $\rightarrow$ random outcome

Bubble: Brocken commons.wikimedia.org
Bursting bubble: wallpaperswide.com

2.1 The mathematics of quantum states

MULT20015

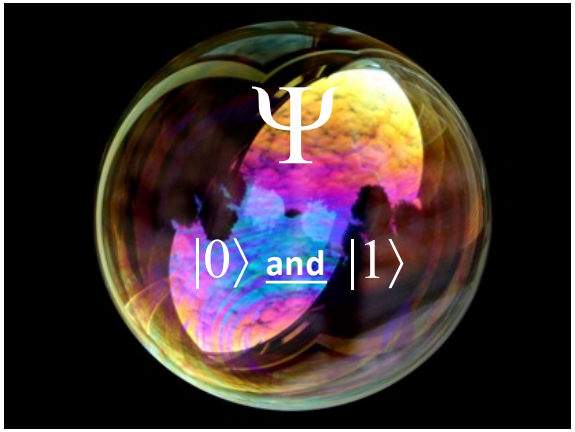
Lecture 2

Quantum bits and probabilities

Prepare

A qubit in a <u>particular</u> quantum superposition, say with more 0 than 1

(as many times as we like)



Measure

quantum superposition collapse

?

or

measurement

repeat the process…

Results

Analyse all outcomes

"0" outcomes

Say 60 times out of 100 trials

Probability: Prob["0"] ~ 60/100 = 0.6

"1" outcomes

Say 40 times out of 100 trials

Probability: Prob["1"] ~ 40/100 = 0.4

Quantum systems are stochastic

Bubble: Brocken commons.wikimedia.org
Bursting bubble: wallpaperswide.com

Qubits – how we describe them mathematically

A qubit, e.g. an electron in an atom

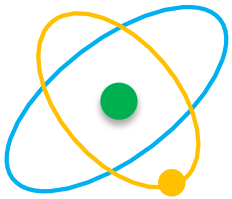
The qubit can be in $|0\rangle$ or $|1\rangle$ or both at the same time.

The overall quantum state is denoted by $|\psi\rangle$

First let's look at the case of no quantum superposition: $|\psi\rangle = |0\rangle$ or $|\psi\rangle = |1\rangle$

i.e. definitive scenarios, one state or the other (no surprises)

$|\psi\rangle = |0\rangle$ The electron is in the <u>ground</u> state



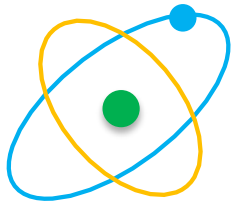
Before: $|\psi\rangle = |0\rangle$

measurement

Collapses to: $|\psi\rangle = |0\rangle$

Outcome always "0" → probability: Prob["0"] = 1.0

$|\psi\rangle = |1\rangle$ The electron is in the <u>excited</u> state



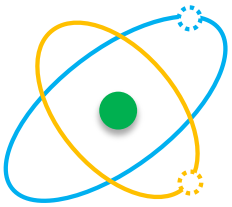
Before: $|\psi\rangle = |1\rangle$

measurement

Collapses to: $|\psi\rangle = |1\rangle$

Outcome always "1" → probability: Prob["1"] = 1.0

Qubit maths – quantum superposition



Qubit in both $|0\rangle$ and $|1\rangle$ states

$\rightarrow$ i.e. $|\psi\rangle$ is a <u>quantum superposition</u> of $|0\rangle$ and $|1\rangle$

Maths: we describe $|\psi\rangle$ as a linear combination of basis states over the linear space $\{|0\rangle,|1\rangle\}$:

$$\boxed{|\psi\rangle = a_0\,|0\rangle + a_1\,|1\rangle}$$

- a_0 and a_1 are the "quantum state amplitudes" (over $\mathbb{C}$ -> complex numbers)

  $\rightarrow$ tell us how much of $|0\rangle$ and $|1\rangle$ are in the superposition $|\psi\rangle$

- modulus squared of the amplitudes, $|a_0|^2$ and $|a_1|^2 \rightarrow$ probabilities of measuring "0" or "1"

- total probability must be 1, so we have the "normalization" condition: $|a_0|^2 + |a_1|^2 = 1$

Let's recap complex numbers…

2.2 Complex numbers and quantum amplitudes

MULT20015

Lecture 2

Complex numbers: basics

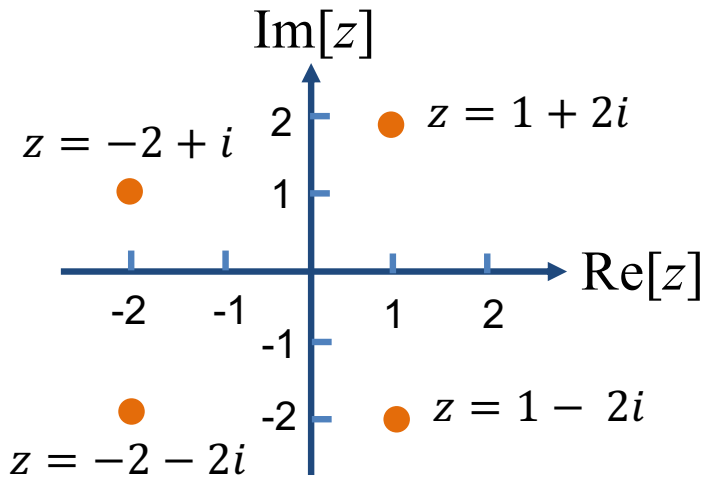
$|\psi\rangle = a_0 |0\rangle + a_1 |1\rangle$ where a_0 and a_1 are complex numbers.

Complex numbers recap: $\boxed{i = \sqrt{-1} \text{ , and so } i^2 = -1}$

$z = x + iy$

"Real" part is denoted by Re[z] = x

"Imaginary" part is denoted by Im[z] = y



Addition: $z_1 + z_2 = (x_1 + iy_1) + (x_2 + iy_2) = (x_1 + x_2) + i(y_1 + y_2)$

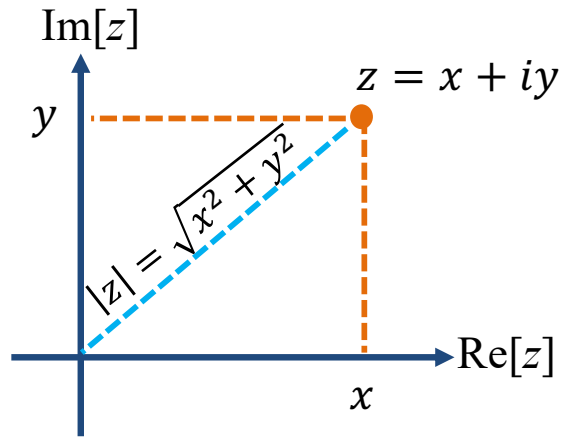
Multiplication: $z_1 z_2 = (x_1 + iy_1)(x_2 + iy_2) = x_1 x_2 - y_1 y_2 + i(x_1 y_2 + y_1 x_2)$

Conjugate: $z = x + iy \rightarrow z^* = x - iy$

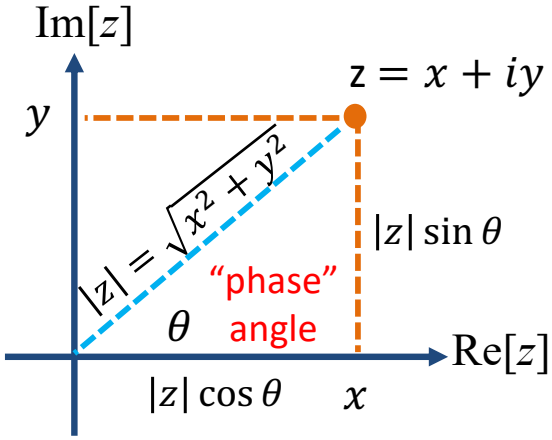
$z\, z^* = (x + iy)(x - iy) = x^2 - ixy + iyx + y^2 = x^2 + y^2$

i.e. $z\, z^* = x^2 + y^2 = |z|^2$ i.e. $|z| = \sqrt{x^2 + y^2}$

$|z|$ is referred to as the "modulus" or "magnitude"



Complex numbers: polar notation



Cartesian: $z = x + iy$ in terms of x=Re[z] and y=Im[z]

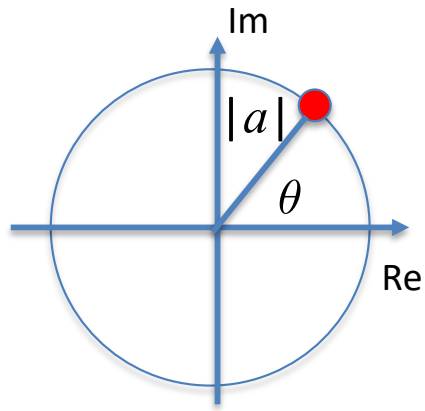
Polar: $z = r\,(\cos\theta + i\sin\theta)$ = $r\,e^{i\theta}$

NB: $e^{i\theta} = \cos\theta + i\sin\theta, r = |z| = \sqrt{x^2 + y^2}$

In the QUI, we use "polar notation". e.g. for amplitudes in the state: $|\psi\rangle = a_0\,|0\rangle + a_1\,|1\rangle$

Ignoring subscripts, consider a complex amplitude *a*

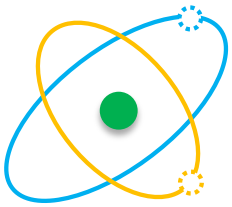
$$a = \mathrm{Re}[a] + i\,\mathrm{Im}[a] = |a|e^{i\theta} \rightarrow \begin{array}{c} |a| = \sqrt{\mathrm{Re}[a]^2 + \mathrm{Im}[a]^2} \\ \theta = \tan^{-1}\left(\mathrm{Im}[a]/\mathrm{Re}[a]\right) \end{array}$$



For each complex amplitude we have a magnitude and phase angle:

$$|\psi\rangle = a_0\,|0\rangle + a_1\,|1\rangle \;\rightarrow |\psi\rangle = |a_0|e^{i\theta_0}\,|0\rangle + |a_1|e^{i\theta_1}\,|1\rangle$$

Continuing: Qubit maths – quantum superposition



Qubit in both $|0\rangle$ and $|1\rangle$ states

$\rightarrow$ i.e. $|\psi\rangle$ is a <u>quantum superposition</u> of $|0\rangle$ and $|1\rangle$

Maths: we describe $|\psi\rangle$ as an "addition" of states over a linear space $\{|0\rangle,|1\rangle\}$:

$$\boxed{|\psi\rangle = a_0\,|0\rangle + a_1\,|1\rangle}$$

- a_0 and a_1 are the "quantum state amplitudes" (complex numbers in general)

  $\rightarrow$ tell us how much of $|0\rangle$ and $|1\rangle$ are in the superposition $|\psi\rangle$

- modulus squared of the amplitudes, $|a_0|^2$ and $|a_1|^2 \rightarrow$ probabilities of measuring "0" or "1"
- total probability must be 1, so we have the "normalization" condition: $|a_0|^2 + |a_1|^2 = 1$

Qubit superposition, amplitudes, and measurement

Back to qubit superposition and measurement.

Amplitudes and probability:

Measurement	Outcome	$\lvert\psi\rangle$ "collapsed" to:	Outcome probability
	"0"	$\lvert\psi\rangle \to \lvert 0\rangle$	Prob["0"] = $\lvert a_0 \rvert^2$
	or		
	"1"	$\lvert\psi\rangle \to \lvert 1\rangle$	Prob["1"] = $\lvert a_1 \rvert^2$

Prepared state $\lvert\psi\rangle$

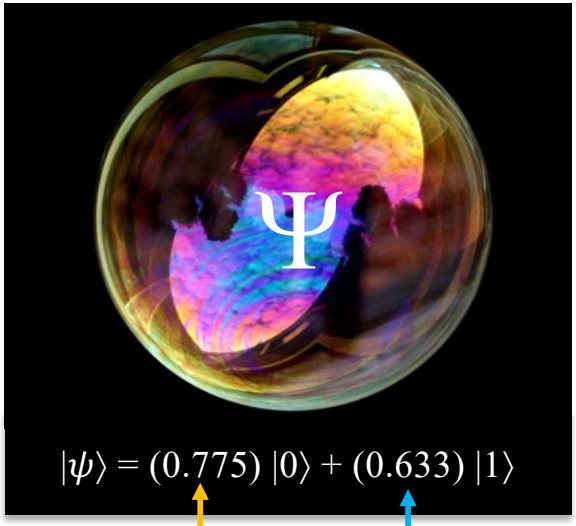
$$\boxed{\lvert\psi\rangle = a_0 \lvert 0\rangle + a_1 \lvert 1\rangle}$$

?

Examples

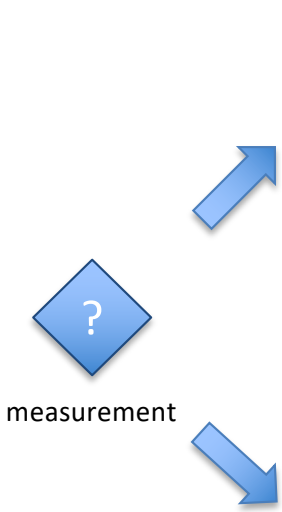
Assume for now the amplitudes are real:

Let's prepare a qubit in that particular "60:40" superposition



$|\psi\rangle = (0.775)\,|0\rangle + (0.633)\,|1\rangle$

a_0 a_1



Measurement

or

The probability of getting a "0" outcome is given by:

Prob["0"] = $(0.775)^2$ = 0.6

The probability of getting a "1" outcome is given by:

Prob["1"] = $(0.633)^2$ = 0.4

NB. In any given prepare-measure shot, the outcome will appear random!

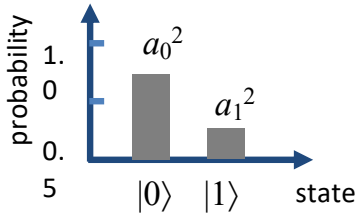
But, by repeating the prepare-measurement process many times

the probabilities emerge from $|\psi\rangle = (0.775)\,|0\rangle + (0.633)\,|1\rangle$

Bubble: Brocken commons.wikimedia.org
Bursting bubble: wallpaperswide.com

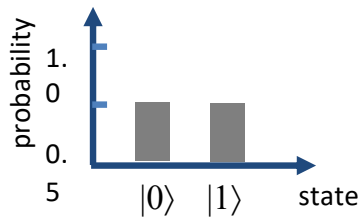
Examples

Assume the amplitudes a_0 and a_1 are real.

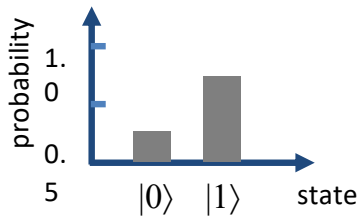
Quantum state (before)	Probability measuring 0	Probability measuring 1	Histogram (many shots)
$\|\psi\rangle = a_0 \|0\rangle + a_1 \|1\rangle$	${a_0}^2$	${a_1}^2$	

${a_0}^2 + {a_1}^2 = 1$

$$|\psi\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle$$



$$|\psi\rangle = \frac{1}{2}|0\rangle + \frac{\sqrt{3}}{2}|1\rangle$$

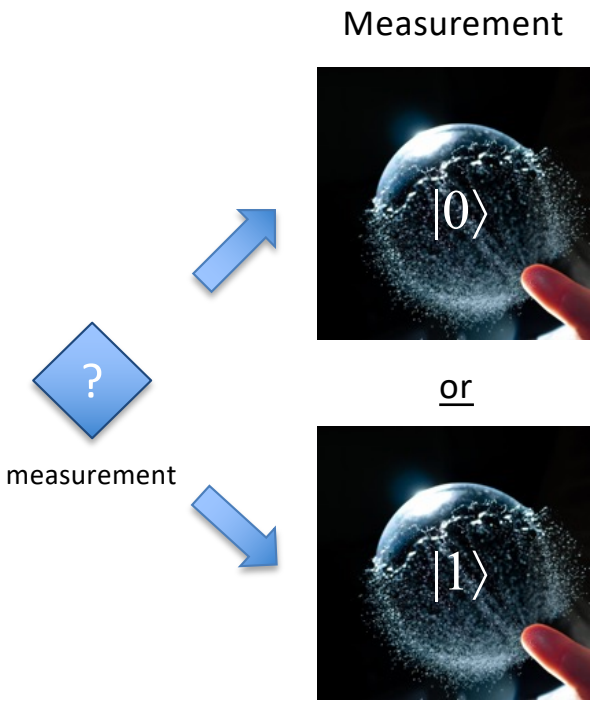


Examples – cont.

Complex amplitudes:

$$|\psi\rangle = 0.775e^{i\pi}\,|0\rangle + 0.633\,e^{-i\pi/3}|1\rangle$$

$|a_0|$ θ_0 $|a_1|$ θ_1



The probability of getting a

"0" outcome is given by:

$$|a_0|^2 = (0.775)^2 = 0.6$$

Prob["0"] = (0.775)^2 = 0.6

The probability of getting a

"1" outcome is given by:

$$|a_0|^2 = (0.633)^2 = 0.4$$

Prob["1"] = (0.633)^2 = 0.4

Bubble: Brocken commons.wikimedia.org
Bursting bubble: wallpaperswide.com

2.3 Basic linear algebra: ket and matrix notation

MULT20015

Lecture 2

Dirac's "ket" notation

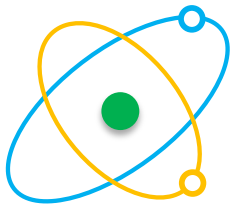
- A lot of quantum mechanics comes down to linear algebra (matrices and vectors), but uses a slightly different notation introduced by Dirac:

$|\psi\rangle$ ← A "ket" is a element of a **linear vector space** over $\mathbb{C}$ which represents *the state* of a qubit.

We write the general state of a qubit in "ket" notation as:

$$|\psi\rangle = a_0 |0\rangle + a_1 |1\rangle$$

Where the quantum amplitudes a_0 and a_1 are complex numbers.



The wave-like attributes of quantum systems are encapsulated by amplitudes represented as complex numbers…

Ignoring subscripts: $a = x + iy$ where x=Re[a] and y=Im[a]

Linear Algebra and Dirac notation

$$|\psi\rangle = a_0 |0\rangle + a_1 |1\rangle$$

For qubits we can use column vectors to represent a convenient basis for kets:

$$|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$|1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

Computational basis states

$$a_0 |0\rangle + a_1 |1\rangle = \begin{bmatrix} a_0 \\ a_1 \end{bmatrix}$$

$$a_0, a_1 \in \mathbb{C}$$

General qubit state
a_0 and a_1 are "amplitudes"

Dual vectors

$\langle\psi|$ A “bra” is a **row vector** .

For a qubit state,

$$|\psi\rangle = \begin{bmatrix} a_0 \\ a_1 \end{bmatrix} \quad a_0, a_1 \in \mathbb{C}$$

we define the corresponding *dual vector* to be:

$$\langle\psi| = \begin{bmatrix} a_0^* & a_1^* \end{bmatrix}$$

Inner Product

$\langle\psi|\phi\rangle$

A "braket" is an **inner product**
(analogous to dot product for vectors in 3D)

For two quantum states $|\psi\rangle = \begin{bmatrix} a \\ b \end{bmatrix}, \quad |\phi\rangle = \begin{bmatrix} c \\ d \end{bmatrix}$

We can define an inner product between them

$$\langle\psi|\phi\rangle \equiv \langle\psi||\phi\rangle$$

$$= \begin{bmatrix} a^* & b^* \end{bmatrix} \begin{bmatrix} c \\ d \end{bmatrix}$$

$$= a^*c + b^*d$$

Orthogonality

Two states are *orthogonal* if their inner product is zero

$$\langle\psi|\phi\rangle = 0$$

"Z-basis" (computational basis)

For $|0\rangle$ and $|1\rangle$

$$\langle 0|1\rangle = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$= 0$$

Computational basis states are orthogonal

"X-basis" (+/- states)

For

$$|+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}$$

$$|-\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}$$

$$\langle +|-\rangle = \frac{1}{2} \begin{bmatrix} 1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$= 0$$

These states are also orthogonal

Outer Product

$|\psi\rangle\langle\phi|$ is an **outer product**

For two quantum states $|\psi\rangle = \begin{bmatrix} a \\ b \end{bmatrix}, \quad |\phi\rangle = \begin{bmatrix} c \\ d \end{bmatrix}$

We can define an outer (tensor) product between them:

$$|\psi\rangle\langle\phi| = \begin{bmatrix} a \\ b \end{bmatrix} \otimes \begin{bmatrix} c^* & d^* \end{bmatrix}$$

$$= \begin{bmatrix} ac^* & ad^* \\ bc^* & bd^* \end{bmatrix}$$

…in case we need it later…

2.4 State representation in the Quantum User Interface (QUI)

MULT20015

Lecture 2

QUI: registration

The QUI is a quantum computer programming and simulation tool developed at the University of Melbourne, used in research and teaching.

The QUI is accessed through a web-based interface.

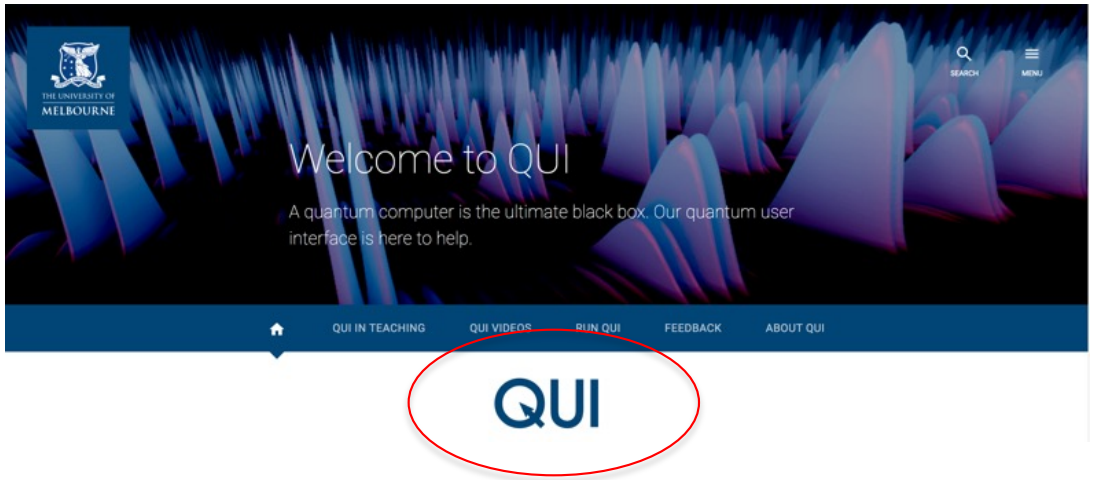
Step 1: Open a web browser (preferably Google Chrome or Firefox), and go to QUIspace.org.

Step 2: Click on QUI

Step 3: Sign up. You will need to create an account to access QUI for the first time.

Follow the steps to create your account (email address and answering a few simple questions).

Step 4: Once you have signed-up, start the QUI!



Prior to practice class 1: register for the QUI as per above (**you must use your UoM email**)

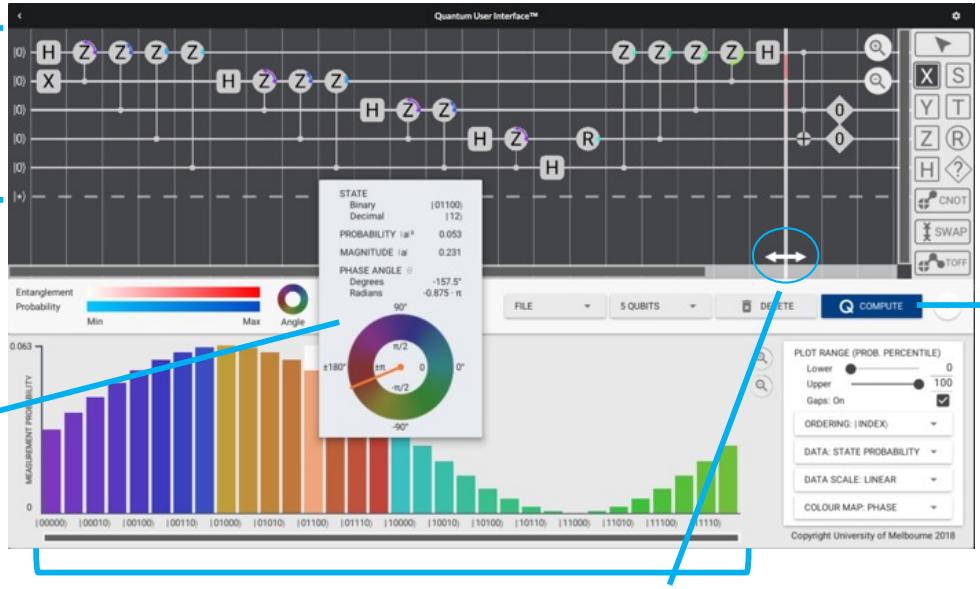
In class 1: switch on expanded capabilities (number of qubits,…)

The QUI structure

- you create a quantum program ("circuit")
- execute ("compute") on the UoM quantum computer simulator
- the results are sent back to your QUI session to display

Program editor – place logic operations on the qubits to create a "circuit"

qubits, initialised in the state $|0\rangle$, with time lines left to right

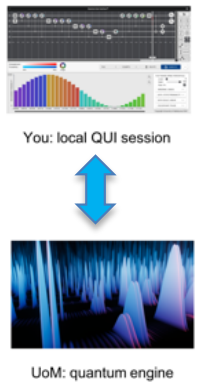


Library of logical operations

Compute – runs program

State information card (SIC) showing more detail from the plot

Plotting controls



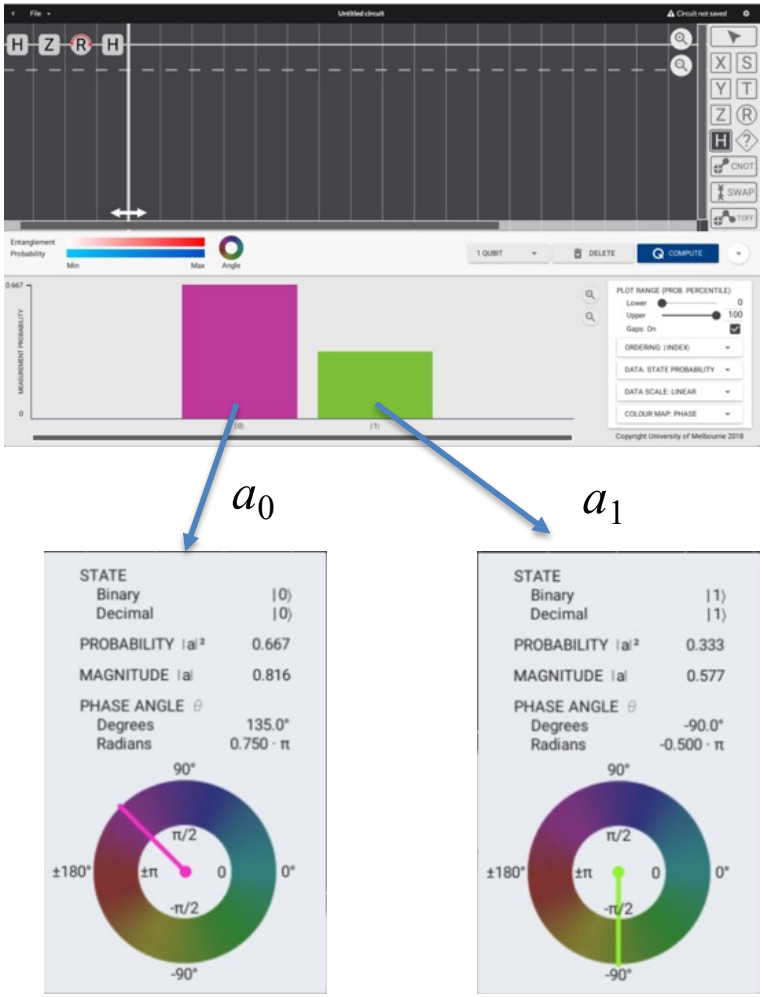
quantum state visualization at slider

quantum computer simulator at UoM

More details: QUIspace.org

Complex numbers in the QUI

Quantum User Interface (QUI) – UoM programming and simulation environment.

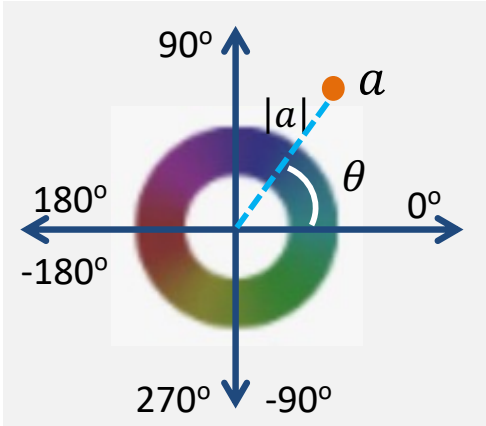


Quantum program in upper panel

Lower panel gives the mathematical representation of the quantum state – i.e. the complex amplitudes (at the time-step corresponding to the vertical slider)

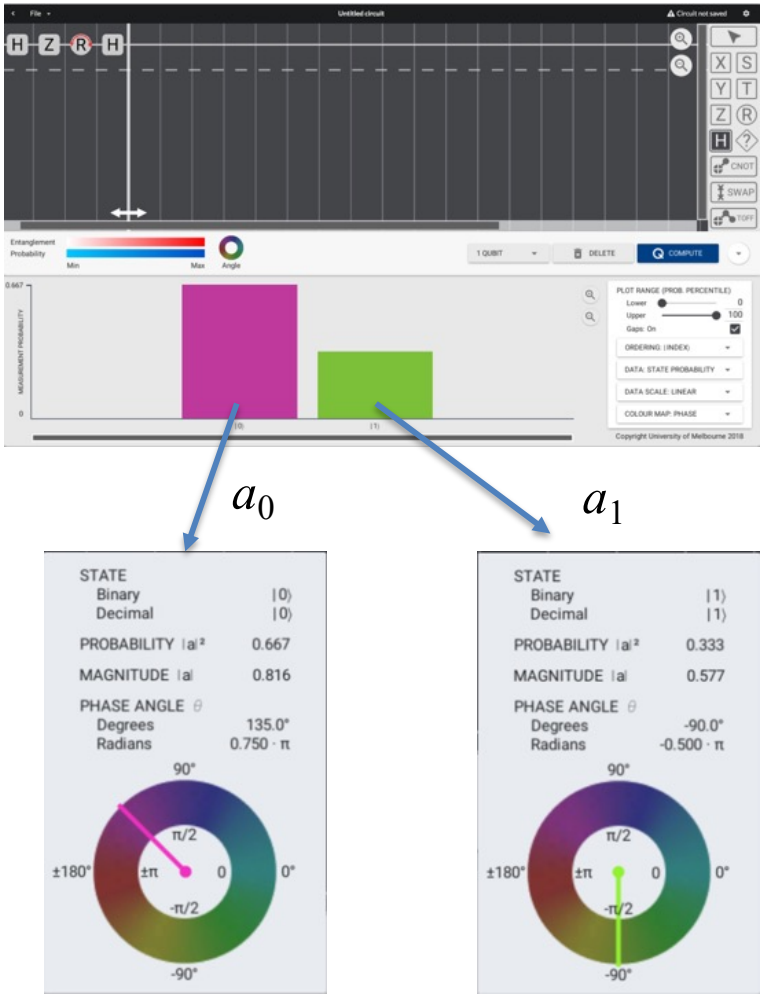
$$|\psi\rangle = a_0\,|0\rangle + a_1\,|1\rangle$$

$$\rightarrow |\psi\rangle = |a_0|e^{i\theta_0}\,|0\rangle + |a_1|e^{i\theta_1}\,|1\rangle$$



“Ket” and “Matrix” representations of quantum states

Consider the following state in the QUI environment:



“ket” notation (amplitudes in QUI polar form)

$$|\psi\rangle = a_0\,|0\rangle + a_1\,|1\rangle$$

$$\rightarrow |\psi\rangle = |a_0|e^{i\theta_0}\,|0\rangle + |a_1|e^{i\theta_1}\,|1\rangle$$

$$a_0 = 0.816 \times e^{i(0.750\,\pi)}$$
$$a_1 = 0.577 \times e^{i(-0.500\,\pi)}$$

“matrix” notation (amplitudes in QUI polar form)

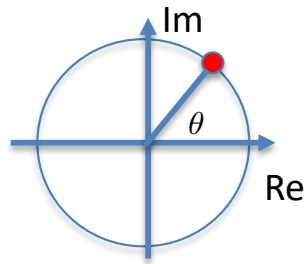
$$a_0|0\rangle + a_1|1\rangle \rightarrow \begin{bmatrix} a_0 \\ a_1 \end{bmatrix}$$

$$a_0|0\rangle + a_1|1\rangle \rightarrow \begin{bmatrix} 0.816 \times e^{i(0.750\,\pi)} \\ 0.577 \times e^{i(-0.500\,\pi)} \end{bmatrix}$$

Amplitudes in the QUI

Quantum systems actually have *wave-like* properties. The complex state amplitudes a_0 and a_1 represent the magnitude and *phase* of this wave.

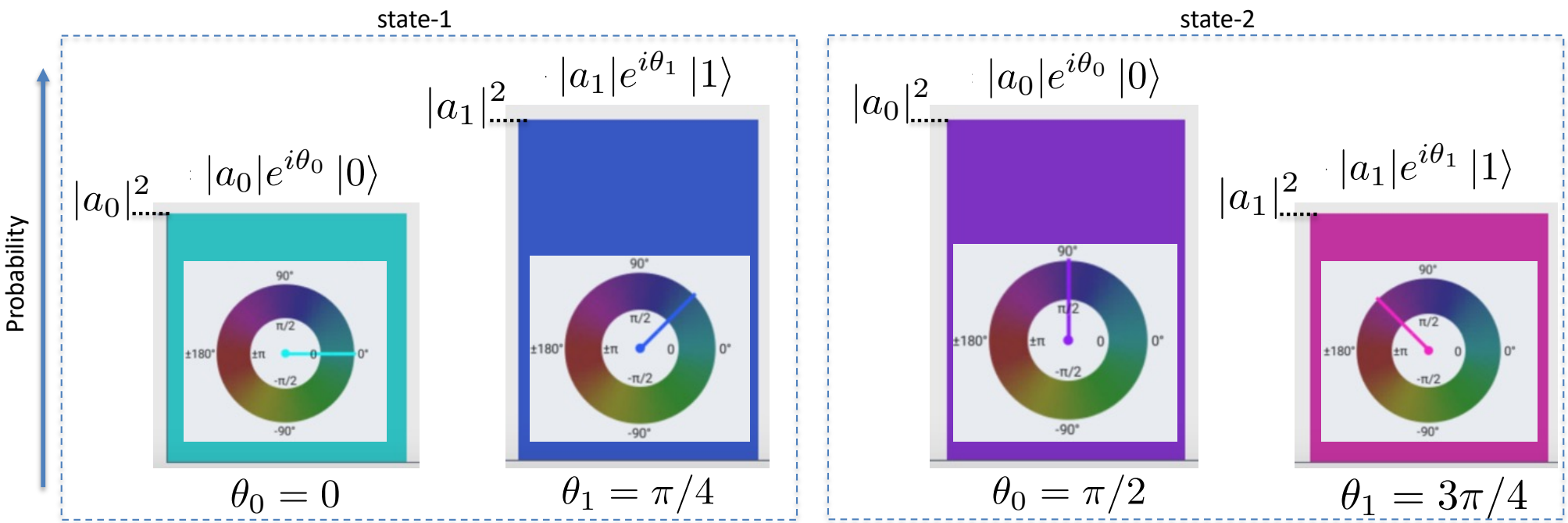
$$|\psi\rangle = a_0\,|0\rangle + a_1\,|1\rangle \qquad \rightarrow |\psi\rangle = |a_0|e^{i\theta_0}\,|0\rangle + |a_1|e^{i\theta_1}\,|1\rangle$$



Recall: $a = \mathrm{Re}[a] + i\,\mathrm{Im}[a] = |a|e^{i\theta} \rightarrow$ $|a| = \sqrt{\mathrm{Re}[a]^2 + \mathrm{Im}[a]^2}$, $\theta = \tan^{-1}\left(\mathrm{Im}[a]/\mathrm{Re}[a]\right)$

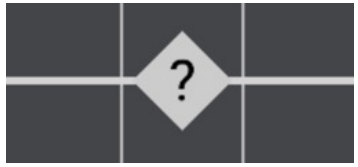
$$e^{i\theta} = \cos\theta + i\sin\theta$$

In the QUI, phase is represented using an angular colour map, and probability by histogram, e.g. consider two different single-qubit states:



QUI: measurement operation

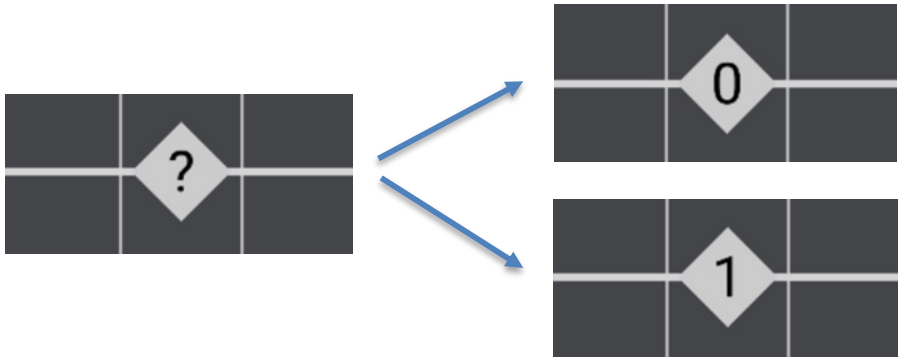
In the QUI the measurement operation looks like this:



By default, measurements are made in the “computational basis” (i.e. 0 or 1).

When you run the circuit, the QUI will randomly select a measurement outcome based on the amplitudes of the state at that point:

$$|\psi\rangle = a_0 |0\rangle + a_1 |1\rangle$$



Prob = $|a_0|^2$

Prob = $|a_1|^2$

In some circuit diagrams notation is: (but we like QUI’s spinning lottery symbol)

Practice classes (aka “tutorials”)

Review and practice lecture concepts/content through QUI exercises.

Practice class sheets -> download from LMS and bring hard copy, and/or access in tutorial

Work through the exercises individually and/or groups

Demonstrator in prac-class there to help -> ask lots of questions!

Week 1

Lecture 1

1.1 A very brief history of computing

1.2 The quantum world and quantum computing

Lecture 2

2.1 The mathematics of quantum states

2.2 Complex numbers and quantum amplitudes

2.3 Basic linear algebra: ket and matrix notation

2.4 State representation in the QUI

Practice class 1

The Quantum User Interface (QUI), lecture 1 & 2 review exercises

Subject outline

Lecture topics (by week)

1 – Introduction to quantum computing and maths basics
2 – Single qubit representations and logic operations
3 – Two qubit states and logic gates
4 – Multi-qubit states and quantum arithmetic
5 – Basic quantum algorithms
6 – Period finding, cryptography and quantum factoring
7 – Shor's algorithm, post-quantum crypto, quantum key distribution
8 – Quantum search algorithms
9 – Grover search applications, optimisation problems
10 – Solving optimisation problems on quantum computers
11 – Applications in quantum machine learning
12 – Real quantum computer devices

Assignment schedule:
#1: Hand out in Week 2
#2: Hand out in Week 8